

Applications of LP-Duality

Max-Flow-Min-Cut Theorem Revisit, von Neumann's Minimax Theorem

Strong Duality Theorem

- Theorem [Strong Duality Theorem]. Let $\mathbf{x}^*$ be the optimal solution to (a) and $\mathbf{y}^*$ be the optimal solution to (b), then $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$.

maximize $\mathbf{c}^T \mathbf{x}$

subject to $A\mathbf{x} \leq \mathbf{b}$ (a)

$\mathbf{x} \geq \mathbf{0}$

minimize $\mathbf{b}^T \mathbf{y}$

subject to $A^T \mathbf{y} \geq \mathbf{c}$ (b)

$\mathbf{y} \geq \mathbf{0}$

Primal feasible

Primal OPT = Dual OPT

Dual feasible



Part I: Max-Flow-Min-Cut Theorem Revisited

The Maximum Flow Problem

- The **maximum flow problem** can be formulated by a linear program.

$$\text{maximize} \quad \sum_{u:(s,u) \in E} f_{su}$$

$$\text{subject to} \quad 0 \leq f_{uv} \leq c_{uv} \quad \forall (u,v) \in E$$

$$\sum_{v:(v,u) \in E} f_{vu} = \sum_{w:(v,w) \in E} f_{uw} \quad \forall u \in V \setminus \{s,t\}$$

Let's Write It in Standard Form

$$\text{maximize} \quad \sum_{u:(s,u) \in E} f_{su}$$

$$\text{subject to} \quad f_{uv} \leq c_{uv} \quad \forall (u, v) \in E$$

$$\sum_{v:(v,u) \in E} f_{vu} - \sum_{w:(u,w) \in E} f_{uw} \leq 0 \quad \forall u \in V \setminus \{s, t\}$$

$$- \sum_{v:(v,u) \in E} f_{vu} + \sum_{w:(u,w) \in E} f_{uw} \leq 0 \quad \forall u \in V \setminus \{s, t\}$$

$$f_{uv} \geq 0 \quad \forall (u, v) \in E$$

We also make it easier

$$\text{maximize} \quad \sum_{u:(s,u) \in E} f_{su}$$

$$\text{subject to} \quad f_{uv} \leq c_{uv} \quad \forall (u, v) \in E \quad \rightarrow \mathbf{y}_{uv}$$

$$\sum_{v:(v,u) \in E} f_{vu} - \sum_{w:(u,w) \in E} f_{uw} = 0 \quad \forall u \in V \setminus \{s, t\} \quad \rightarrow \mathbf{z}_u$$

$$f_{uv} \geq 0 \quad \forall (u, v) \in E$$

Compute Its Dual Program

$$\begin{array}{llll} \text{minimize} & \sum_{(u,v) \in E} c_{uv} y_{uv} & & \\ \text{subject to} & y_{su} + z_u \geq 1 & \forall u: (s, u) \in E & \\ & y_{vt} - z_v \geq 0 & \forall v: (v, t) \in E & \\ & y_{uv} - z_u + z_v \geq 0 & \forall (u, v) \in E, u \neq s, v \neq t & \\ & y_{uv} \geq 0 & \forall (u, v) \in E & \end{array}$$

- We aim to show the LP above describes the min-cut problem.
- Let OPT_{dual} be its optimal objective value. We need to show OPT_{dual} is the size of the min-cut.

Some Intuitions

$$\text{minimize} \quad \sum_{(u,v) \in E} c_{uv} y_{uv}$$

$$\text{subject to} \quad y_{su} + z_u \geq 1$$

$$\forall u: (s, u) \in E$$

$$y_{vt} - z_v \geq 0$$

$$\forall v: (v, t) \in E$$

$$y_{uv} - z_u + z_v \geq 0$$

$$\forall (u, v) \in E, u \neq s, v \neq t$$

$$y_{uv} \geq 0$$

$$\forall (u, v) \in E$$

- y_{uv} describes if edge (u, v) is cut:

$$y_{uv} = \begin{cases} 1 & \text{if } (u, v) \text{ is cut} \\ 0 & \text{otherwise} \end{cases}$$

- z_u describes u 's "side":

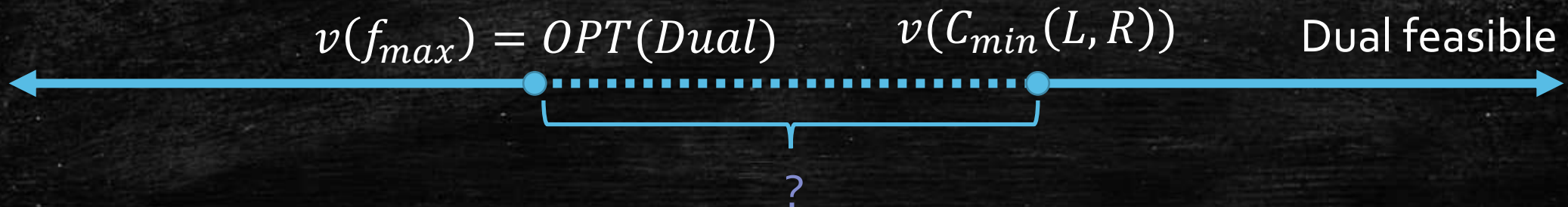
$$z_u = \begin{cases} 1 & \text{if } u \text{ is on the } s - \text{side} \\ 0 & \text{if } u \text{ is on the } t - \text{side} \end{cases}$$

Do you have any ideas?

Feasible Cut is only a subset of
Feasible Dual.

Strong Duality

- A straight-forward idea
- Max Flow: $v(f_{max}) = OPT(Primal) = OPT(Dual)$
- Min Cut: $v(C_{min}(L, R)) = OPT(Dual)$?
- Is that straight-forward?
- Min Cut: need integer solution!
- $OPT(Dual) \geq OPT(Cut\ IP) \geq v(C_{min}(L, R))$



Strong LP-Duality $\Rightarrow$ Max-Flow-Min-Cut

Use Strong Duality Theorem to prove max-flow-min-cut theorem:

- ✓ Step 1: Write down the LP for max-flow problem.
- ✓ Step 2: Show that the dual program describes **the fractional version of** the min-cut problem.
- Step 3: Show that the dual program always have **integral optimum**.
 - So that the dual optimum is indeed the size of min-cut.
- ✓ Step 4: apply Strong Duality Theorem to show max-flow = min-cut

A General Question

maximize $\mathbf{c}^T \mathbf{x}$

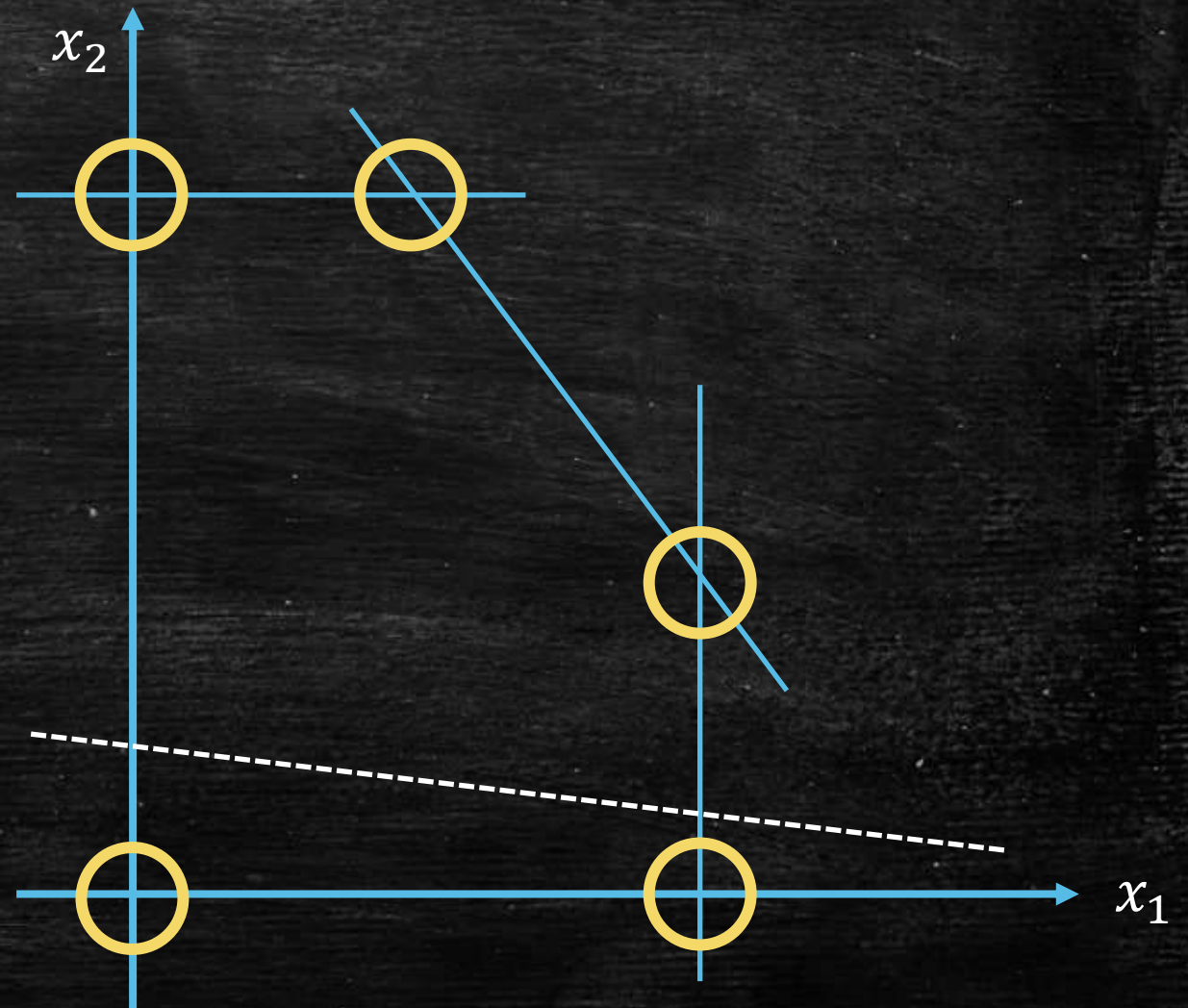
subject to $A\mathbf{x} \leq \mathbf{b}$

$\mathbf{x} \geq \mathbf{0}$

- When the LP has optimal **integral solution**?
- Thinking about the simplex method?

A simple observation

- If all vertices are integral,
- Then the optimal point must be integral.
- Next question: when will it be true?
- They are some solutions of



Totally Unimodular Matrix

- **Definition.** A matrix A is **totally unimodular** if every square submatrix has determinant 0, 1 or -1 .
- **Theorem.** If $A \in \mathbb{R}^{m \times n}$ is totally unimodular and $\mathbf{b}$ is an integer vector, then the polytope $P = \{\mathbf{x}: A\mathbf{x} \leq \mathbf{b}\}$ has integer vertices.
- A Proof Sketch.
 - If $\mathbf{v} \in \mathbb{R}^n$ is a vertex of P . Then there exists an invertible square submatrix A' of A such that $A'\mathbf{v} = \mathbf{b}'$ for some sub-vector $\mathbf{b}'$ of $\mathbf{b}$.
 - By Cramer's Rule, we have $v_i = \frac{\det(A'_i|\mathbf{b}')}{\det(A'_i)}$, where $(A'_i|\mathbf{b}')$ is the matrix with i -th column replaced by $\mathbf{b}'$.
 - $\det(A'_i) = \pm 1$ and $\det(A'_i|\mathbf{b}') \in \mathbb{Z}$. Thus, $\mathbf{v}$ is integral.

Corollary on Integrality of LP

- **Theorem.** If $A \in \mathbb{R}^{m \times n}$ is totally unimodular and $\mathbf{b}$ is an integer vector, then the polytope $P = \{\mathbf{x}: A\mathbf{x} \leq \mathbf{b}\}$ has integer vertices.
- Since there always exists optimum at a **vertex** of the feasible region of LP, we have the following corollary.
- **Corollary.** If A is unimodular, then the optimal solution to LP (a) is integral when $\mathbf{b}$ is integral, and the optimal solution to LP (b) is integral when $\mathbf{c}$ is integral.

$$\begin{array}{ll} \text{maximize} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array} \quad (\text{a})$$

$$\begin{array}{ll} \text{minimize} & \mathbf{b}^T \mathbf{y} \\ \text{subject to} & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0} \end{array} \quad (\text{b})$$

Proving Integrality of y_{uv}, z_u

$$\text{minimize} \quad \sum_{(u,v) \in E} c_{uv} y_{uv}$$

$$\text{subject to} \quad y_{su} + z_u \geq 1$$

$$\forall u: (s, u) \in E$$

$$y_{vt} - z_v \geq 0$$

$$\forall v: (v, t) \in E$$

$$y_{uv} - z_u + z_v \geq 0$$

$$\forall (u, v) \in E, u \neq s, v \neq t$$

$$y_{uv} \geq 0$$

$$\forall (u, v) \in E$$

- Now, we show that the matrix describing the first three rows of the constraints is totally unimodular.

Proving Integrality of y_{uv}, z_u

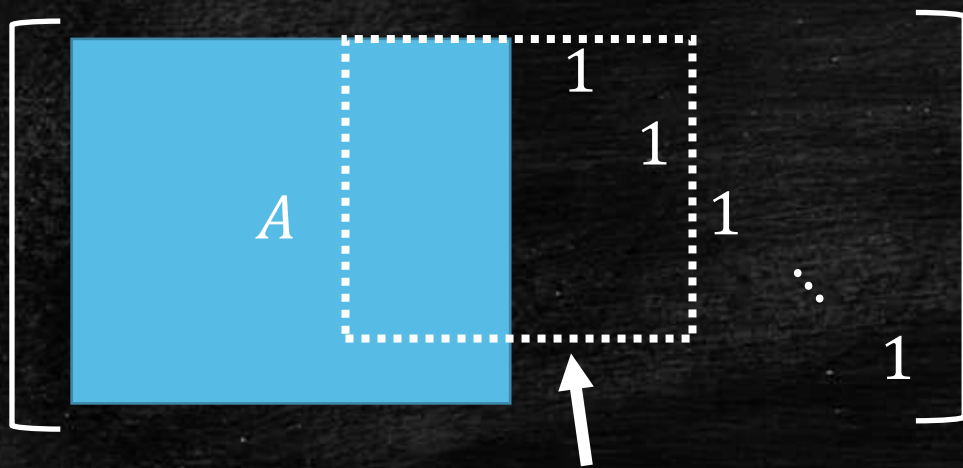
- The matrix can be written below:

$$\begin{array}{c}
 \begin{array}{c} |E| \\ \underbrace{\hspace{1.5cm}} \end{array} \quad \begin{array}{c} |V| - 2 \\ \underbrace{\hspace{2.5cm}} \end{array} \\
 \begin{array}{c} \underbrace{\hspace{1.5cm}} \\ |E| \end{array} \left[\begin{array}{cc} \begin{array}{c} |E| \times |E| \\ \text{identity} \\ \text{matrix} \end{array} & \begin{array}{cc} u & v \\ 1 & \\ -1 & 1 \\ & -1 \end{array} \end{array} \right] \begin{array}{c} (s, u) \\ (u, v) \\ (v, t) \end{array} \\
 \begin{array}{cc} Y & Z \end{array}
 \end{array}$$

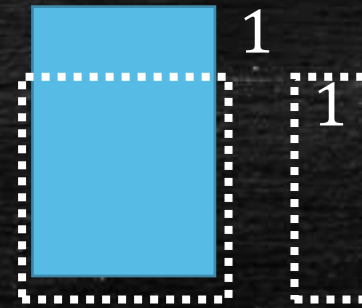
- Let the matrix be $[Y \ Z]$. Y is the identity matrix. We only need to show Z is totally unimodular.

Some Simple Observations

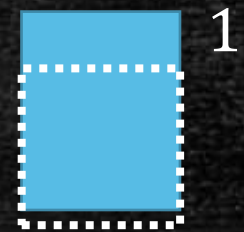
- If A is totally unimodular, then so are A^T , $[I \ A]$, $[A \ I]$, $\begin{bmatrix} I \\ A \end{bmatrix}$, and $\begin{bmatrix} A \\ I \end{bmatrix}$. If any of A^T , $[I \ A]$, $[A \ I]$, $\begin{bmatrix} I \\ A \end{bmatrix}$, and $\begin{bmatrix} A \\ I \end{bmatrix}$ is totally unimodular, then so is A .
- Proof. Just expand the determinant and you will see it...
- The determinant of $[A \ I]$ equals to ± 1 times the determinant of some square submatrix of A .



Consider this submatrix



Expand on this column



Expand on this column

Proving Integrality of y_{uv}, z_u

- The matrix can be written below:

$$\begin{array}{c}
 \begin{array}{c} |E| \left\{ \begin{array}{c} \overbrace{\begin{array}{cc} u & v \end{array}}^{|V|} \\ \left[\begin{array}{cc} 1 & \\ -1 & 1 \\ & -1 \end{array} \right] \end{array} \right. \begin{array}{l} (s, u) \\ (u, v) \\ (v, t) \end{array}
 \end{array}
 \end{array}$$

Z

- Let the matrix be $[Y \ Z]$. Y is the identity matrix. We only need to show Z is totally unimodular.

Proving Z is totally unimodular by Induction...

- Base Step: Each cell of Z belongs to $\{0, 1, -1\}$.
- Inductive Step: Suppose every $k \times k$ submatrix of Z has determinant belongs to $\{0, 1, -1\}$. Consider any $(k + 1) \times (k + 1)$ submatrix Z' .
- Case 1: If a row of Z' is all-zero, then $\det(Z') = 0$.
- Case 2: If a row of Z' contains only one non-zero entry, then $\det(Z')$ equals to ± 1 times the determinant of a $k \times k$ submatrix. $\det(Z') \in \{0, 1, -1\}$ by induction hypothesis.
- Case 3: If every row of Z' has two non-zero entries (one of them is -1 and the other is 1), then $\det(Z') = 0$:
 - Adding all the column vectors, we get a zero vector.

Proving Integrality of y_{uv}, z_u

$$\text{minimize} \quad \sum_{(u,v) \in E} c_{uv} y_{uv}$$

$$\text{subject to} \quad y_{su} + z_u \geq 1$$

$$\forall u: (s, u) \in E$$

$$y_{vt} - z_v \geq 0$$

$$\forall v: (v, t) \in E$$

$$y_{uv} - z_u + z_v \geq 0$$

$$\forall (u, v) \in E, u \neq s, v \neq t$$

$$y_{uv} \geq 0$$

$$\forall (u, v) \in E$$

- Now, we conclude that there exists an optimal solution with $y_{uv}, z_u \in \mathbb{Z}$.

Collect what we have.

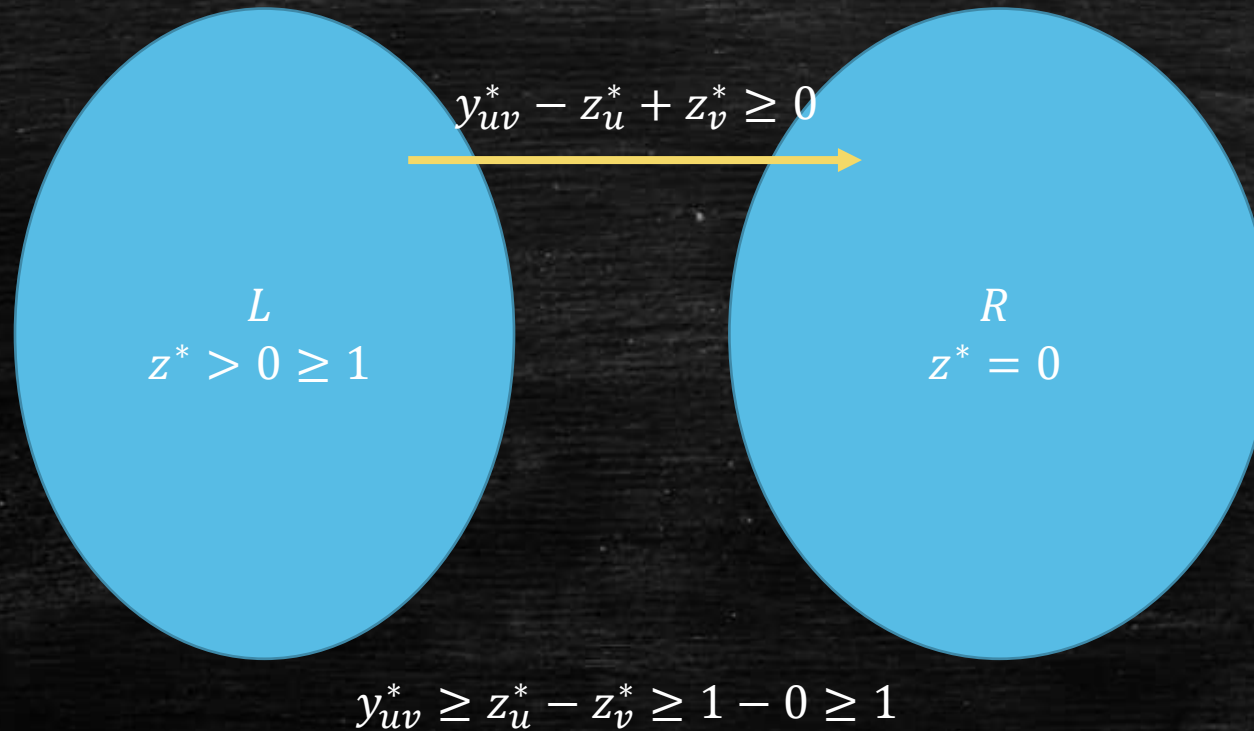
- $OPT(Dual) = OPT(Primal) = v(f_{max})$
- $OPT(Dual)$ can be achieved by integral y^* and z^* .
- Are we done?
- We need to show $OPT(Dual)$ can be achieved by a cut!



- y_{uv} describes if edge (u, v) is cut: ?
$$y_{uv} = \begin{cases} 1 & \text{if } (u, v) \text{ is cut} \\ 0 & \text{otherwise} \end{cases}$$
- z_u describes u 's "side":
$$z_u = \begin{cases} 1 & \text{if } u \text{ is on the } s - \text{side} \\ 0 & \text{if } u \text{ is on the } t - \text{side} \end{cases}$$

Some Intuitions

- Let $L = \{v \mid z_v^* > 0\} + \{s\}$, and consider the cut $\{L, R = V - L\}$.
 - We make $s \in L$ and $t \notin R$.



We are done!

$$\text{minimize} \quad \sum_{(u,v) \in E} c_{uv} y_{uv}$$

- We find the cut $c(L, R)$.
 - $C(L, R) = \sum_{(u,v) \in \text{out}(L)} c(u, v) \leq \sum_{(u,v) \in \text{out}(L)} y_{uv}^* c(u, v) \leq OPT(Dual)$
- Also, because $OPT(Dual) \leq C(L, R)$
 - $C(L, R)$ is a choice of feasible dual.
- We have:
- $MinCut = C(L, R) = OPT(Dual) = OPT(Primal) = MaxFlow$

A Framework for Proving Theorems Using Strong Duality

- Write down the primal and dual LPs.
- Justify that the primal and dual LPs describe the corresponding problems.
- If the problem described is discrete, prove that the corresponding LP always gives integral solution.
 - Total Unimodularity
- Apply strong duality theorem.

Revisiting Integrality Theorem for Max-Flow

- **Theorem.** If the capacities are all integers, then there exists an integral maximum flow.
- We have seen that " A " in the LP is totally unimodular
 - For dual program, we have proved A^T is totally unimodular.
- If all c_{uv} are integers, then vector " b " in the LP is integral, and the LP has an integral optimal solution.

$$\text{maximize} \quad \sum_{u:(s,u) \in E} f_{su}$$

$$\text{subject to} \quad f_{uv} \leq c_{uv}$$

$$\sum_{v:(v,u) \in E} f_{vu} - \sum_{w:(u,w) \in E} f_{uw} \leq 0$$

$$- \sum_{v:(v,u) \in E} f_{vu} + \sum_{w:(u,w) \in E} f_{uw} \leq 0$$

$$f_{uv} \geq 0$$

Exercise: König's Theorem

Maximum Bipartite Matching = Minimum Vertex Cover

Part II: von Neumann's Minimax Theorem

Zero-Sum Game

- Two players: A and B
- Each player has a set of **actions** that (s)he can play.
 - Set of actions A can play: $\mathbf{a} = \{a_1, a_2, \dots, a_m\}$
 - Set of actions B can play: $\mathbf{b} = \{b_1, b_2, \dots, b_n\}$
- For each pair of actions (a_i, b_j) that two players play, a **utility** is assigned to each player: $u_A(a_i, b_j), u_B(a_i, b_j)$.
- A game is a zero-sum game if $\forall x_i, y_j: u_A(a_i, b_j) + u_B(a_i, b_j) = 0$.
- **Payoff Matrix** $G \in \mathbb{R}^{m \times n}$, where $G_{i,j}$ is the **utility gain** for A , or the **utility loss** for B , when (a_i, b_j) is played.

Example

- The payoff matrix for the Rock-Scissors-Paper game:

		Player <i>B</i>		
		Rock	Scissors	Paper
Player <i>A</i>	Rock	0	1	-1
	Scissors	-1	0	1
	Paper	1	-1	0

Strategy

- Set of actions A can play: $\mathbf{a} = \{a_1, a_2, \dots, a_m\}$
- A **strategy** for A is a probability distribution of $\mathbf{x}$.
- A **pure strategy** specifies one of $a_1, a_2, \dots, a_m$ with probability 1.
 - In other words, a pure strategy is an action.
- Otherwise, it is a **mixed strategy**.
 - In other words, a mixed strategy specify at least two actions with non-zero probability.
- Fix A 's strategy, the **best response** for B is the strategy that maximizes B 's utility.

Rock-Scissors-Paper Example

- A plays $(R, S, P) = (1, 0, 0)$:
 - It is a pure strategy that always plays “rock”.
 - The best response for B is $(0, 0, 1)$, with utility 1.
- A plays $(R, S, P) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$:
 - It is a mixed strategy.
 - The best response for B is $(0, 0, 1)$, with expected utility
$$\frac{1}{2} \times 1 + \frac{1}{4} \times -1 + \frac{1}{4} \times 0 = \frac{1}{4}.$$
- A plays $(R, S, P) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$:
 - It is a mixed strategy.
 - Any strategy for B , pure or mixed, is a best response, with expected utility 0.

Expected Utility

- Let $\mathbf{x} = \{x_1, \dots, x_m\}$ and $\mathbf{y} = \{y_1, \dots, y_n\}$ be the strategies played by the two players.

- The expected utility for Player A is

$$U_A(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top G \mathbf{y} = \sum_{i,j} G_{i,j} x_i y_j$$

- The expected utility for Player B is

$$U_B(\mathbf{x}, \mathbf{y}) = -\mathbf{x}^\top G \mathbf{y} = -\sum_{i,j} G_{i,j} x_i y_j$$

Does it matter who chooses strategy first?

Rock-Scissors-Paper: $G = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$

- Suppose A chooses a strategy first.
 - Given that B will always play the best response
 - The optimal strategy for A is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
 - Expected utility for both players is 0
- Suppose B chooses a strategy first.
 - Similar analysis, expected utility for both players is 0
- Same outcome regardless who chooses strategy first.
- Does it always hold for any zero-sum game?
- Yes! This is **von Neumann's Minimax Theorem**.

Minimax Theorem

- Suppose A chooses strategy first. Knowing that B will play the best response, A will choose an optimal strategy $\mathbf{x}$ that maximizes his/her utility:

B plays the best response given A 's strategy $\mathbf{x}$.

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \sum_{i,j} G_{i,j} x_i y_j$$

Given B plays the best response, A choose a strategy maximizing the utility.

- Suppose B chooses strategy first. Similarly, the utility for A is

$$\min_{\mathbf{y}} \max_{\mathbf{x}} \sum_{i,j} G_{i,j} x_i y_j$$

Minimax Theorem

- Minimax Theorem:

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \sum_{i,j} G_{i,j} x_i y_j = \min_{\mathbf{y}} \max_{\mathbf{x}} \sum_{i,j} G_{i,j} x_i y_j$$

- Who chooses strategy first doesn't matter!

Pure Strategy Best Response

- **Lemma.** Fix A 's strategy $\mathbf{x} = \{x_1, \dots, x_m\}$, there exists a best response for B that is a pure strategy.

- Proof. Let $\mathbf{y} = \{y_1, \dots, y_n\}$ be B 's strategy.

- The utility for B is given by

$$-y_1 \sum_{i=1}^m G_{i,1} x_i - y_2 \sum_{i=1}^m G_{i,2} x_i - \dots - y_n \sum_{i=1}^m G_{i,n} x_i$$

- Clearly, this is maximized if we set $y_i = 1$ where y_i has smallest coefficient.

LP formulation

- The lemma implies

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \sum_{i,j} G_{i,j} x_i y_j = \max_{\mathbf{x}} \min_{j=1,\dots,n} \sum_i G_{i,j} x_i$$

- Let z be the utility for Player A. The following LP formulates the max-min expression:

maximize z

subject to $\sum_i G_{i,j} x_i \geq z \quad \forall j = 1, \dots, n$

$$x_1 + \dots + x_m = 1$$

$$x_1, \dots, x_m \geq 0$$

Its dual program is...

maximize z

subject to $z - \sum_i G_{i,j} x_i \leq 0 \quad \forall j = 1, \dots, n \rightarrow \mathbf{y_1 \sim y_n}$

$x_1 + \dots + x_m = 1 \quad \rightarrow \mathbf{w}$

$x_1, \dots, x_m \geq 0$

- Dual Objective: w
- Constraints
 - For z : we do not have $z \geq 0$ so, z 's coefficient should be exactly 1.
 - For x_i : The coefficient should at least 0.

Simplify it, we get...

minimize w

subject to $\sum_j G_{i,j} y_j \leq w \quad \forall i = 1, \dots, m$

$$y_1 + \dots + y_n = 1$$

$$y_1, \dots, y_n \geq 0$$

- This is exactly

$$\min_y \max_x \sum_{i,j} G_{i,j} x_i y_j = \min_y \max_{i=1,\dots,m} \sum_{i,j} G_{i,j} y_j$$

- Strong duality theorem $\Rightarrow$ Minimax Theorem.